VICTORY 2027

Sense Organs

LECTURE-3 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Eye Defects
2.


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

RATTON KO KARO BYE BYE !!
EASY CONCEPTS SE KARO HELLO HI !!

With – Uppu Ma'am



Question

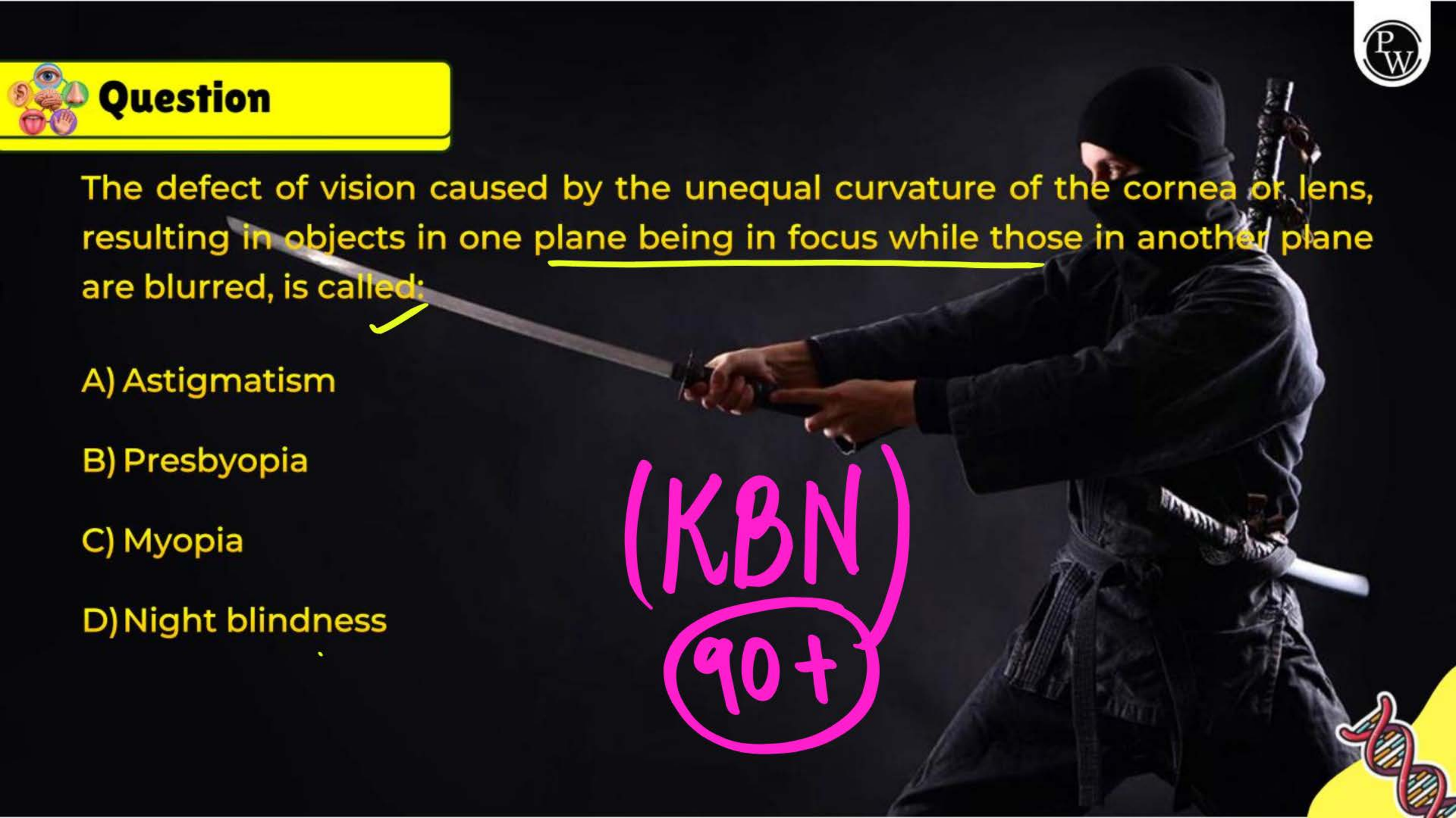
The defect of vision caused by the unequal curvature of the cornea or lens, resulting in objects in one plane being in focus while those in another plane are blurred, is called:

A) Astigmatism

B) Presbyopia

C) Myopia

D) Night blindness



Question

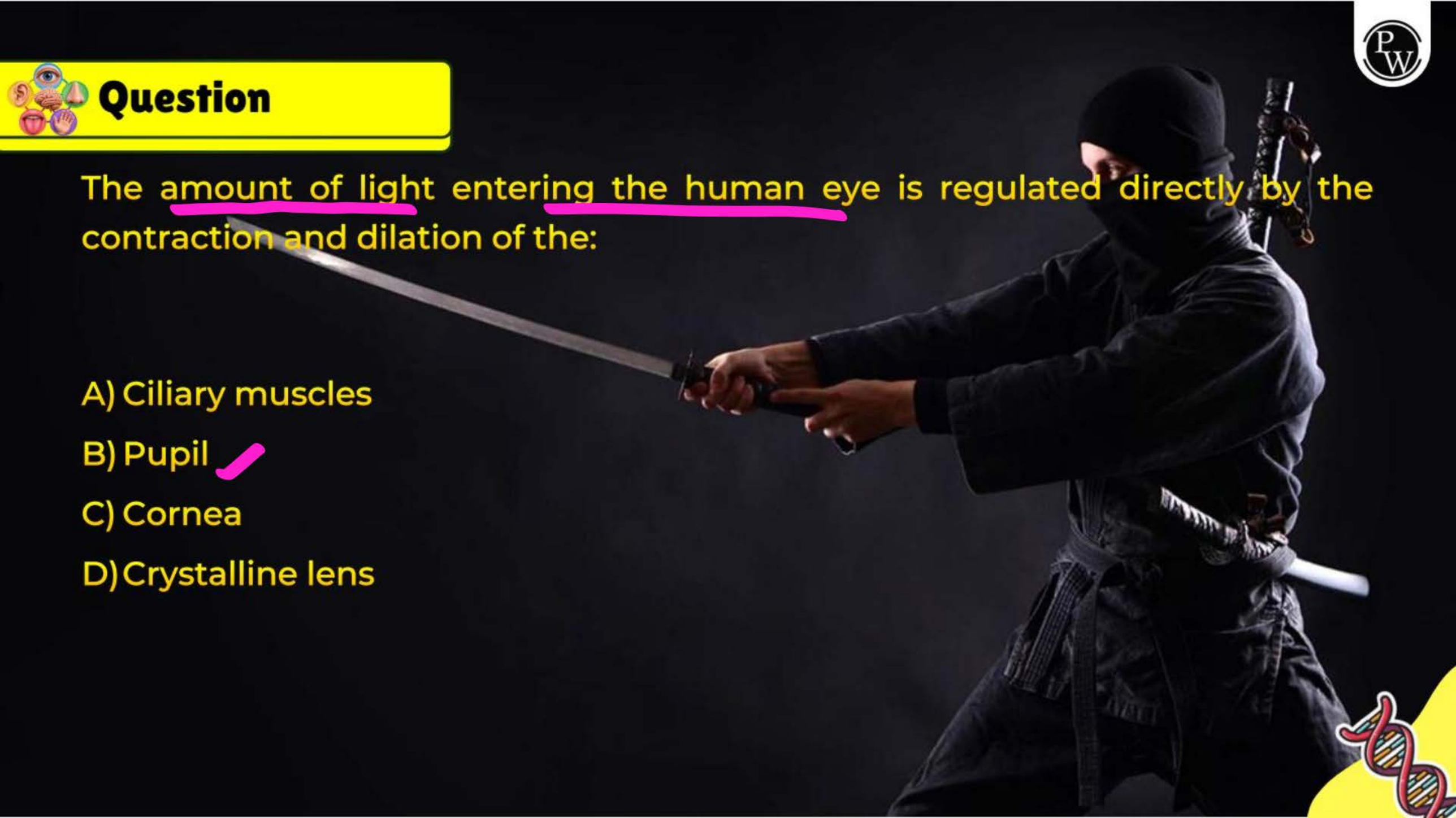
The amount of light entering the human eye is regulated directly by the contraction and dilation of the:

A) Ciliary muscles

B) Pupil

C) Cornea

D) Crystalline lens



Question

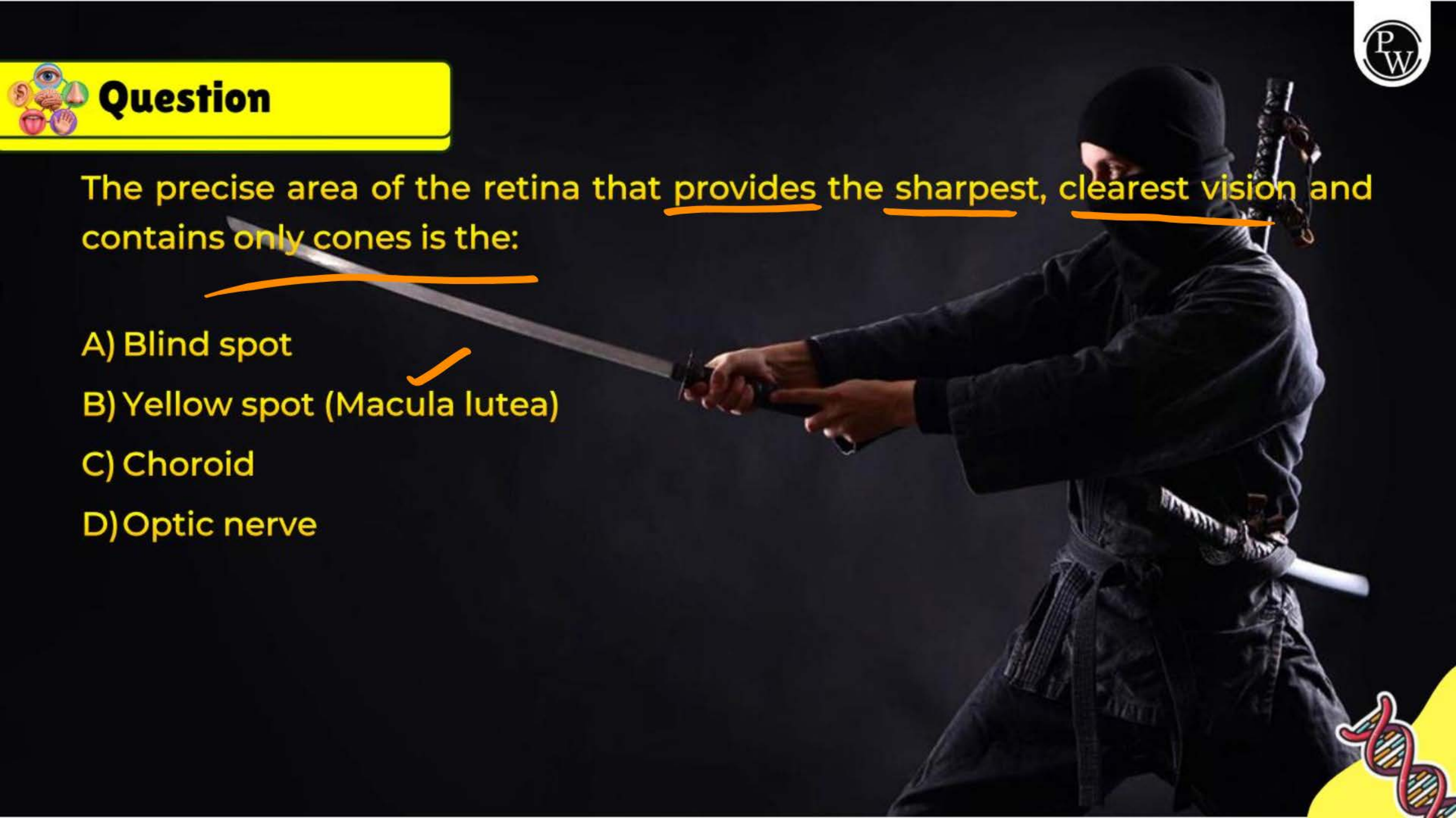
The precise area of the retina that provides the sharpest, clearest vision and contains only cones is the:

A) Blind spot

B) Yellow spot (Macula lutea)

C) Choroid

D) Optic nerve



Question

Which type of photoreceptor cells are highly sensitive to dim light but cannot distinguish colors?

A) Cone cells

B) Rod cells

C) Epithelial cells

D) Ganglion cells



Question

Which type of photoreceptor cells are highly sensitive to dim light but cannot distinguish colors?

A) Cone cells

B) Rod cells

C) Epithelial cells

D) Ganglion cells



front.

The liquid present in the anterior chamber between the cornea and the lens is called:

A) Vitreous humor

B) Cerebrospinal fluid

C) Perilymph

D) Aqueous humor

HOW DO WE SEE?

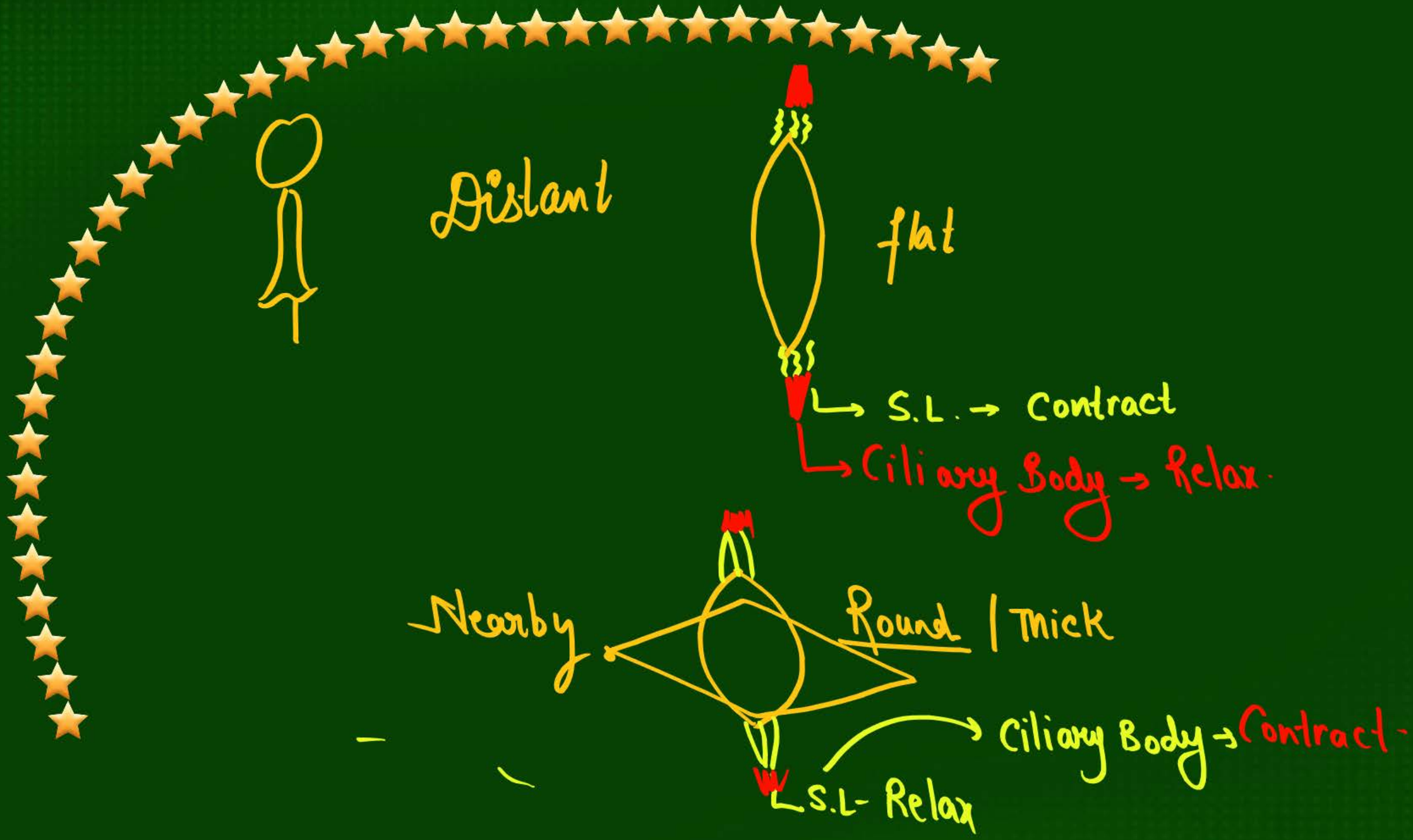
When a person moves from a dark room into bright sunlight what happens to their pupil and iris?

A) Circular muscles relax, pupil dilates

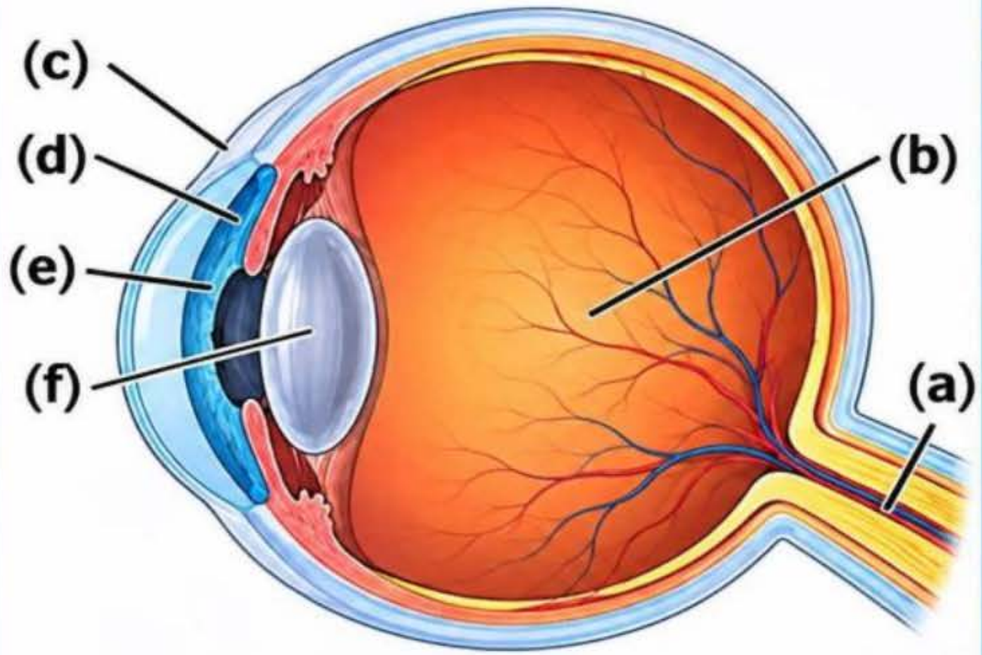
B) Radial muscles contract, pupil dilates

C) Circular muscles contract, pupil constricts

D) Radial muscles contract, pupil constricts


Distant
flat
S.L. → Contract
Ciliary Body → Relax
Nearby
Round / Thick
Ciliary Body → Contract
S.L - Relax

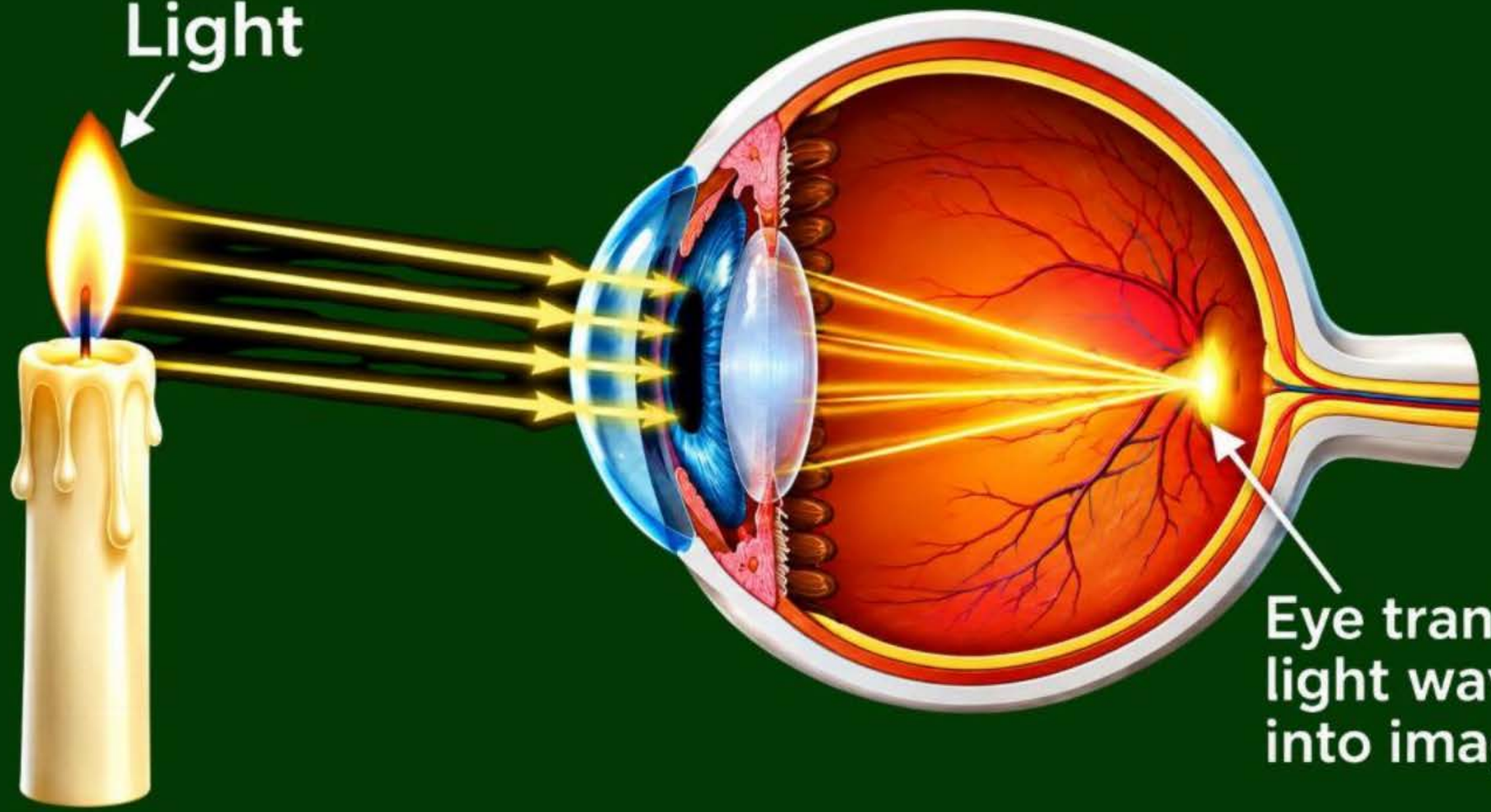
10. Given below *is* a cross section of the **human eye**.
Match the structures marked **(a) to (e)** with their correct functions:



Functions	
1.	Protects retina
2.	Regulates the size of the pupil
3.	Alters the shape of the lens
4.	Keeps the lens moist
5.	Transmits nerve impulses to brain
6.	Holds the lens in position

H.W
PYQ.

HOW DO WE SEE?





HOW DO WE SEE?

Accommodation (viewing objects in sharp focus).

PYQ.

The process of focusing the eye to see objects at different distances is called accommodation.

For distant vision, the lens is more flattened or thinner.

For near vision (nearer than 6 metres), **the lens becomes more convex or rounded. These changes in the shape of the lens is brought about by the ciliary muscles.**

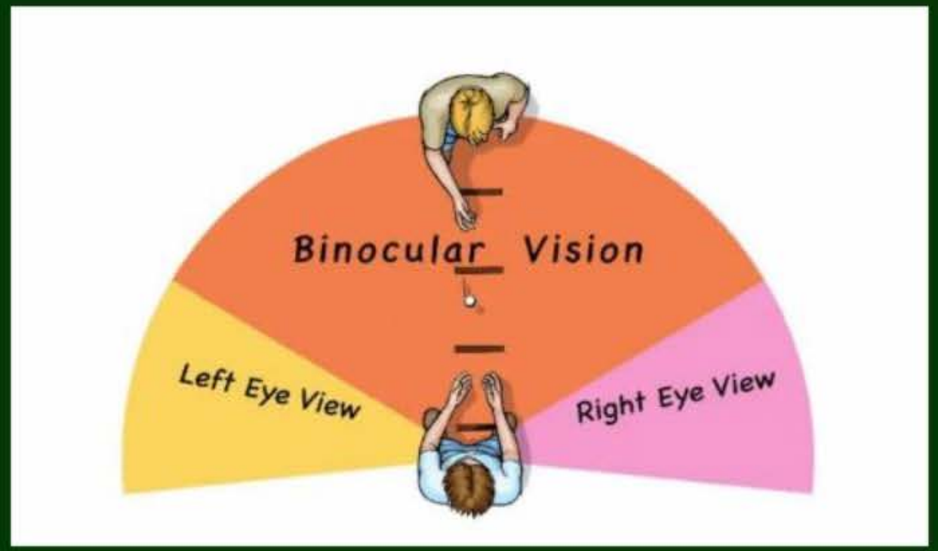


HOW DO WE SEE?

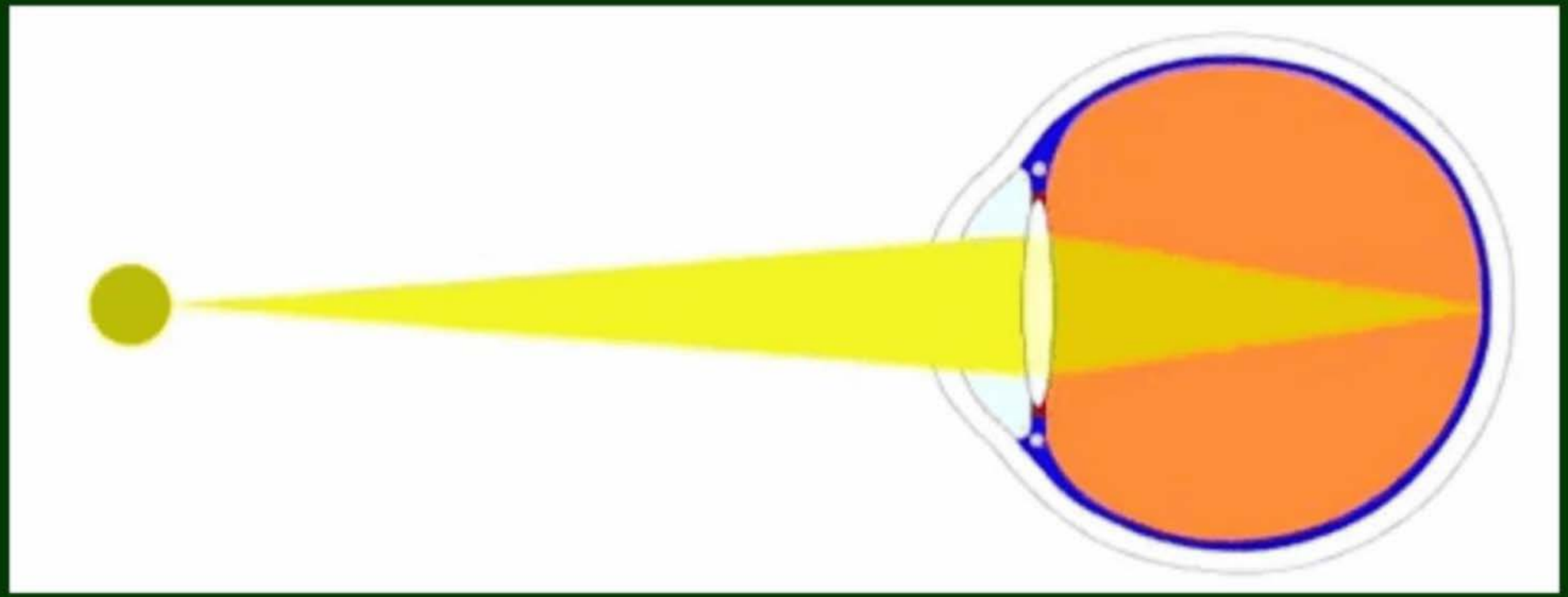
Definition is imp.

Binocular vision/Stereoscopic vision

Binocular vision is the ability of both eyes to work together as a team to create a single, clear, three-dimensional image in the brain







LIGHT AND DARK ADAPTATION

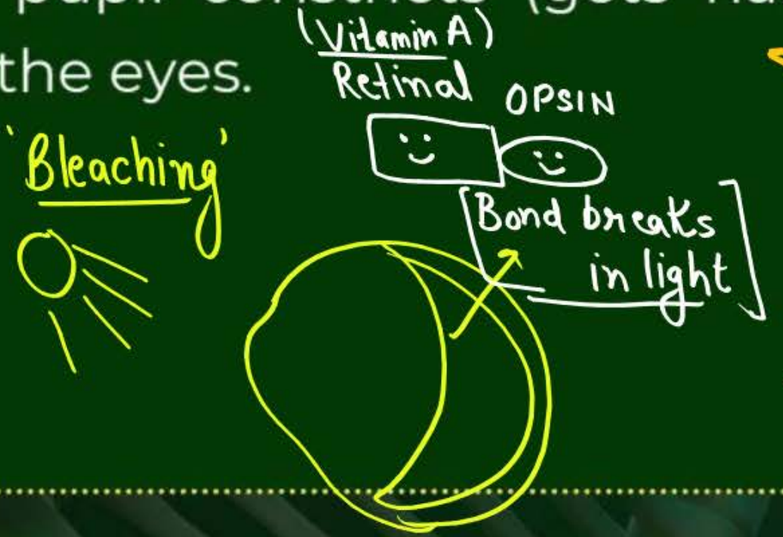
LIGHT ADAPTATION

When a person with dark adapted eyes moves to a brightly lit area, as in coming out of a cinema hall after the noon show, he experiences a dazzling effect for a short period. This is called light adaptation.

Reason of light adaptation:

(a) The visual purple of the rods is bleached (→ Break), reducing their sensitivity, and

(b) The pupil constricts (gets narrower), to reduce the amount of light entering the eyes.





ROD [pigment → Rhodopsin / Visual purple]

CONE [pigment → Iodopsin / Visual violet]

- Chloropsin
- Cyanopsin
- Erythropsin

} Read ONLY

LIGHT AND DARK ADAPTATION

lens → Sus. Ligament वाला main naam Ciliary Body
IRIS → pupil का size.

DARK ADAPTATION:

When you pass from a brightly lit area to a dark room (such as a cinema hall), you experience difficulty in seeing objects for a short while. Slowly, your vision is improved. This improvement is called dark adaptation.

REASON OF DARK ADAPTATION:

Bond reformed b/w Retinal & Opsin

(a) Regeneration of the visual purple or rhodopsin, the pigment of the rods, which was earlier broken down due to bright light, and

(b) Dilation of the pupil permitting more light to enter the eyes.



HOW DO WE SEE?

The four major steps in seeing an object are as follows:

(1) **Entry of light rays :** Light rays from the object enter the eyes through the transparent structures (conjunctiva, cornea, aqueous humour, lens, vitreous humour).

(2) **Focusing of image :** First, the curvature of the cornea converges the light rays to some extent and the lens converges them further to form an image on the retina. The image on the retina is inverted and real.

(3) **Transmission of nerve impulses from retina to brain :** due to chemical changes in cones and rod cells , nerve impulse is sent to brain through optic nerve

(4) **Interpretation by the brain :** image formed in the eye is inverted.



COMMON DEFECTS OF THE EYE : MYOPIA

PYQ v.imp

Short/near sightedness

Near objects can be seen clearly while the distant objects appear blurred.

The image of distant objects is formed in front of the retina.

Reasons for myopia :

(i) the eye-ball is lengthened from front to back

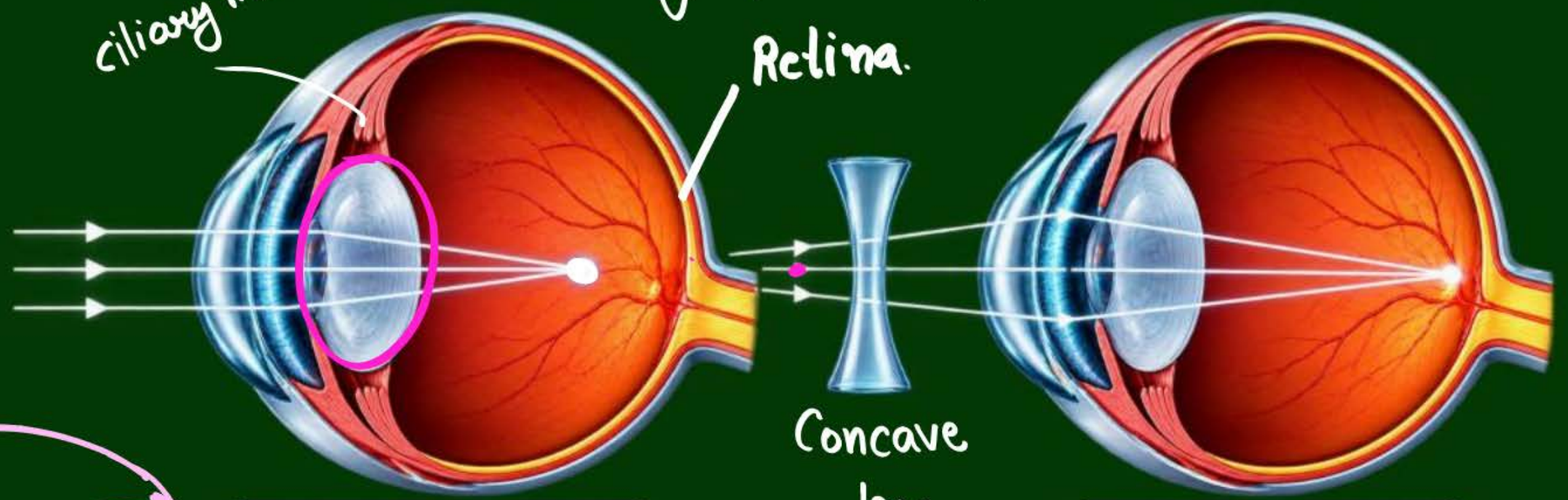
(ii) the lens is too curved (even both reasons may occur together.

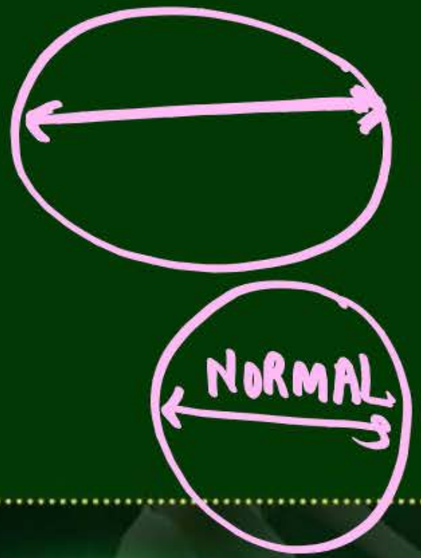
Corrections for myopia

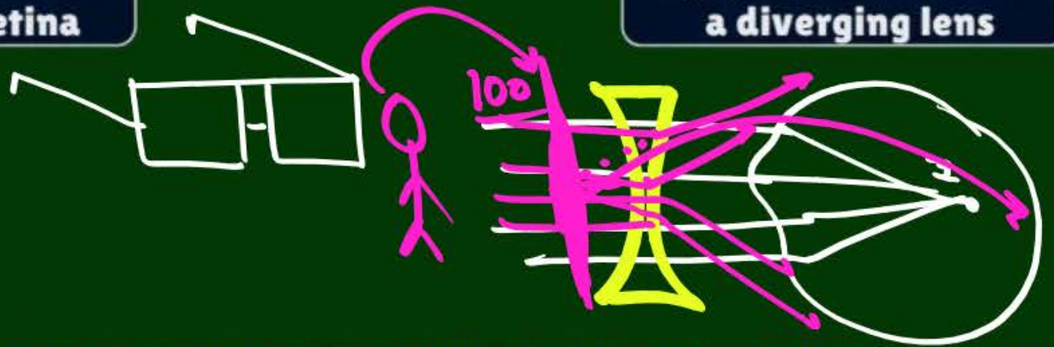
This defect can be corrected by suitable concave (diverging) lens which causes the light rays to diverge before they strike the lens of the eye.

(power of the glasses used is mentioned in plus “-").

CORRECTION OF MYOPIA







Mayonnaise. → MYOPIA.

makes

Elon → Elongated.

Thick → Lens Thick.

went

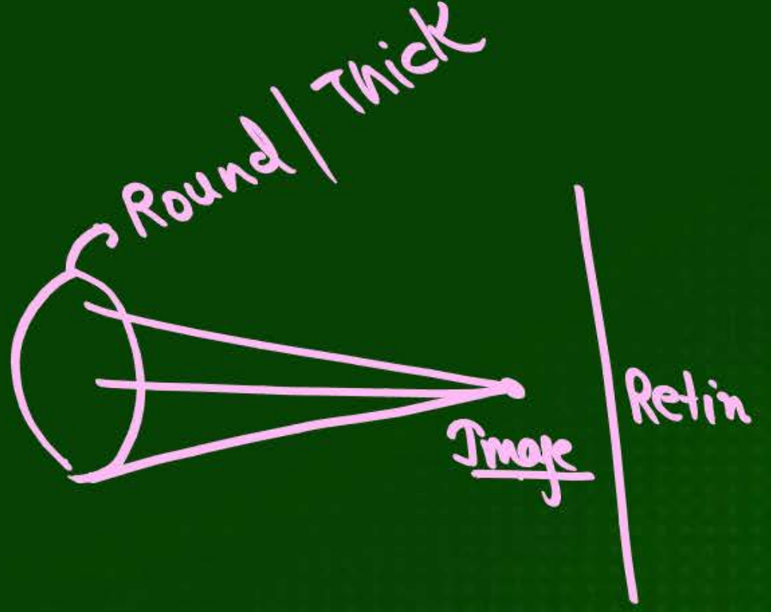
Cave → Concave.

By → Before

Ting → Retina

See.

दूर



CORRECTION OF HYPERMETROPIA

opp. of MYOPIA
Far sightedness

Far or long-sightedness (Hyperopia, Old Term Hypermetropia)

A condition in which there is a difficulty in seeing near objects.

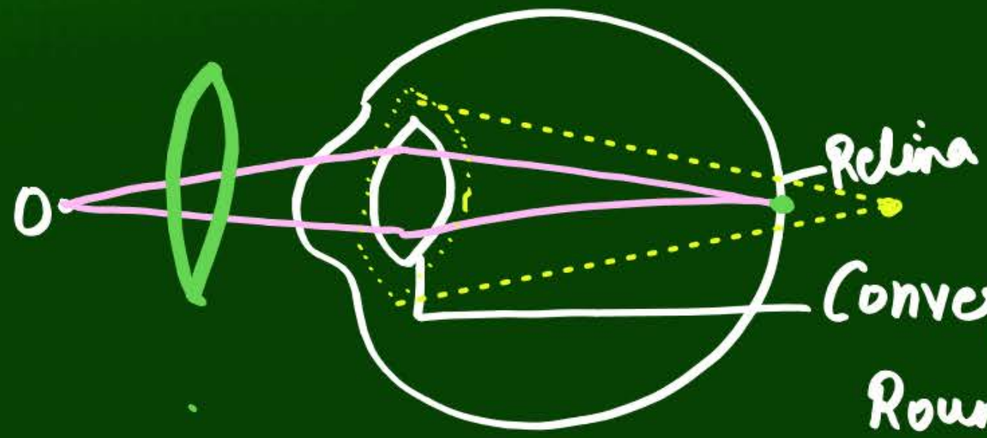
The image of near object falls behind the retina.

Reasons for hyperopia :

Shortening of the eyeball from front to back the lens is too flat. (or thin)

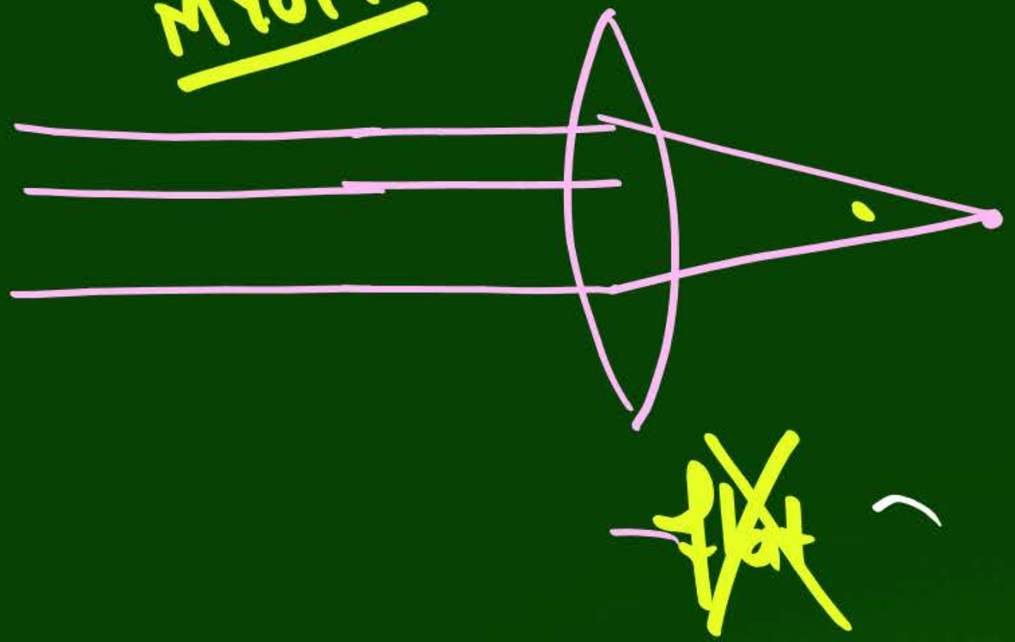
Correction of hyperopia: A convex (converging) Tens is required to correct it (power of the glasses used is mentioned in plus "+").

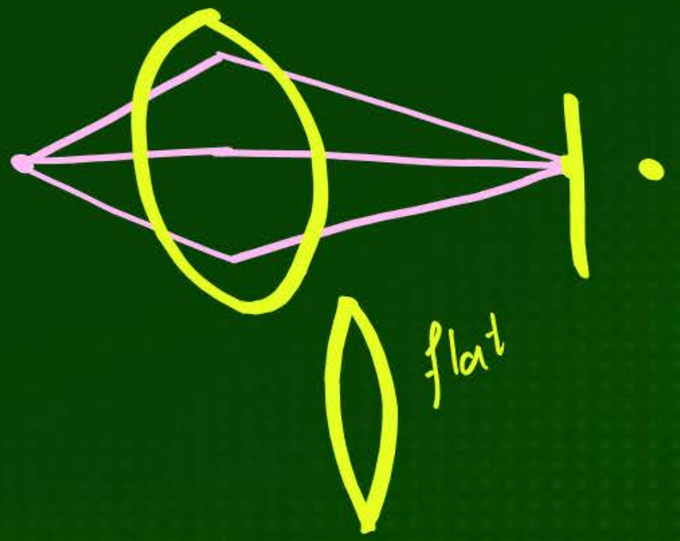




NORMAL

MYOPIA





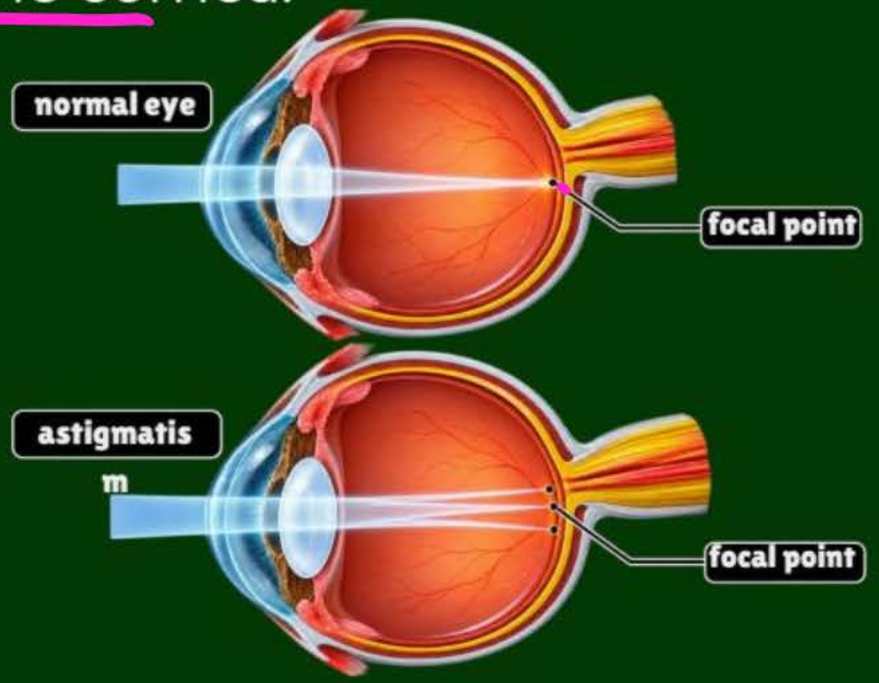
ASTIGMATISM

It is a defect in which some parts of the object are seen in focus while others are blurred.

It arises due to the uneven curvature of the cornea.

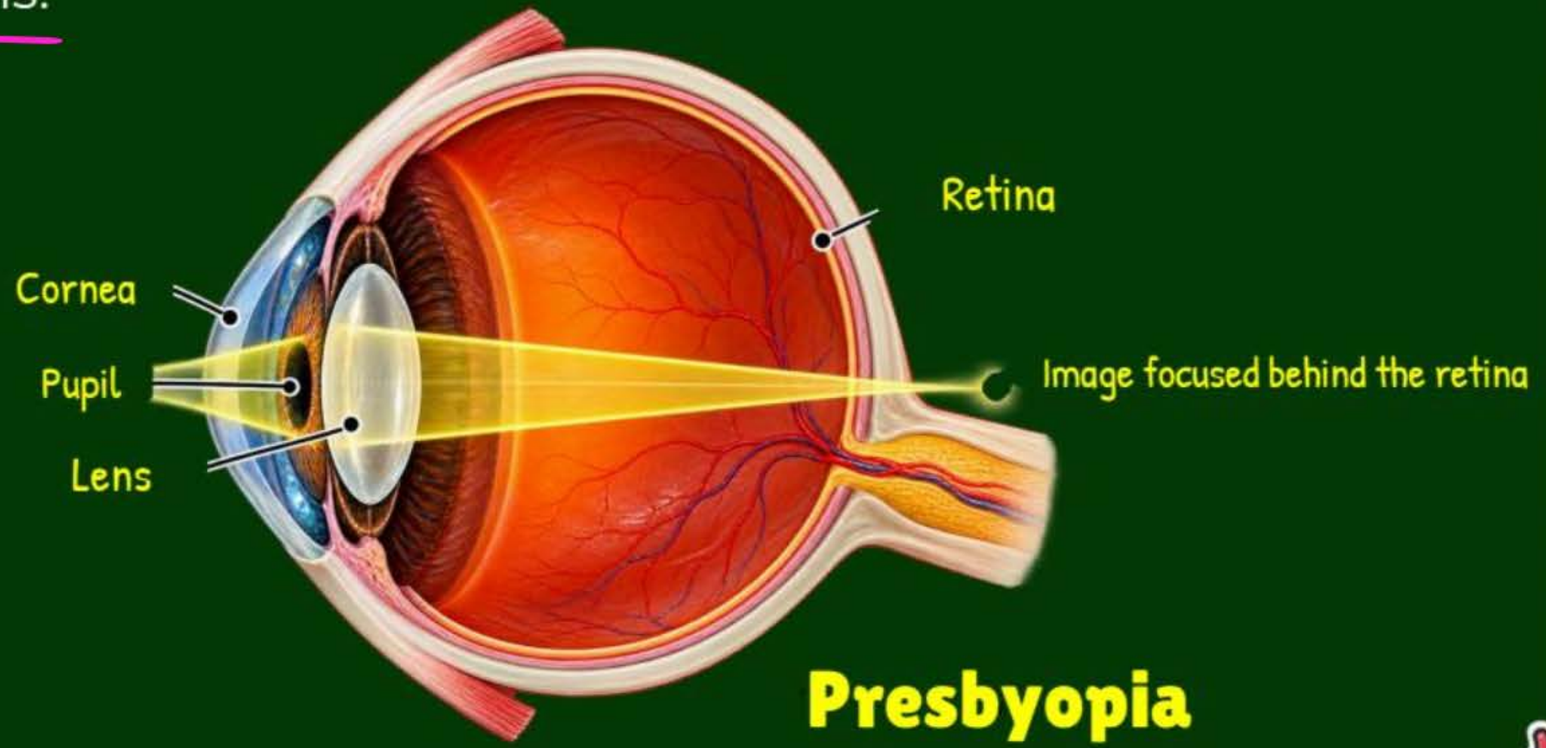
This is corrected by cylindrical lenses.





PRESBYOPIA

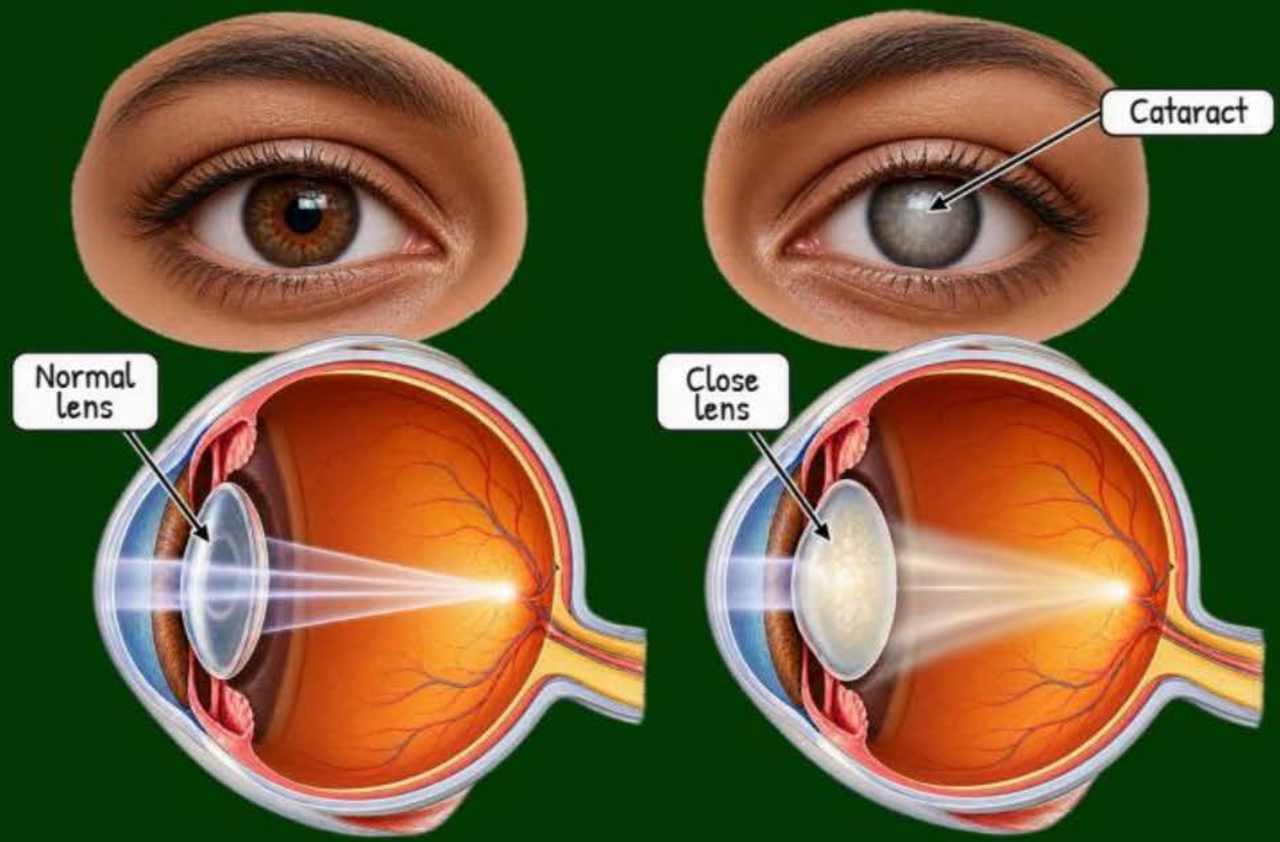
- It is a condition affecting older people who cannot see near objects clearly.
- Their lens loses flexibility resulting in a kind of farsightedness. This again is corrected by a convex lens.



CATARACT

It is a condition in which the lens turns opaque, and the vision is cut down even to total blindness. [Old age,

It can be corrected by surgically removing the lens, and by using spectacles with highly convex lenses



NIGHT-BLINDNESS

It is a condition in which a person feels difficulty in seeing in dim light as during the night.

Rhodopsin Retinal + OPSIN

This is due to non-formation of the pigment visual purple of the rods.

Only rods function in dim light and in the absence of the pigment, they cannot function. This is usually due to the deficiency of vitamin A which is required for the synthesis of the pigment.

Retinal

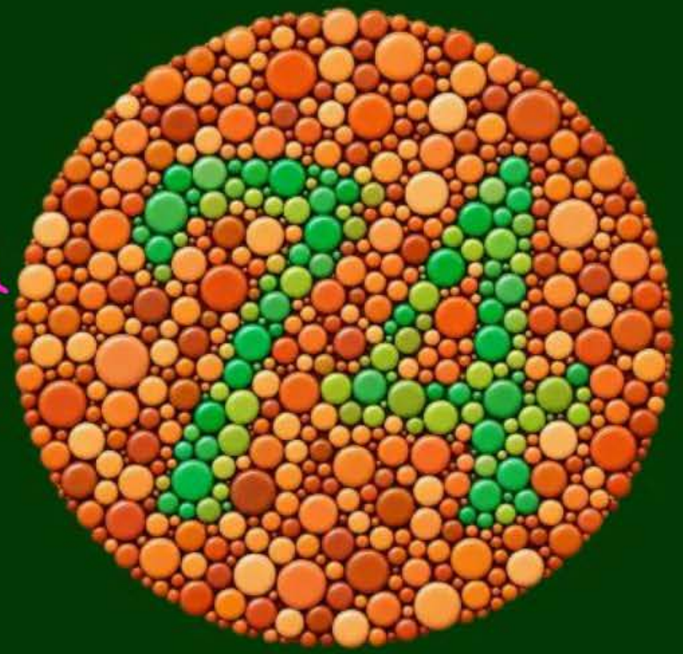


Night blindness Normal Vision

COLOUR BLINDNESS

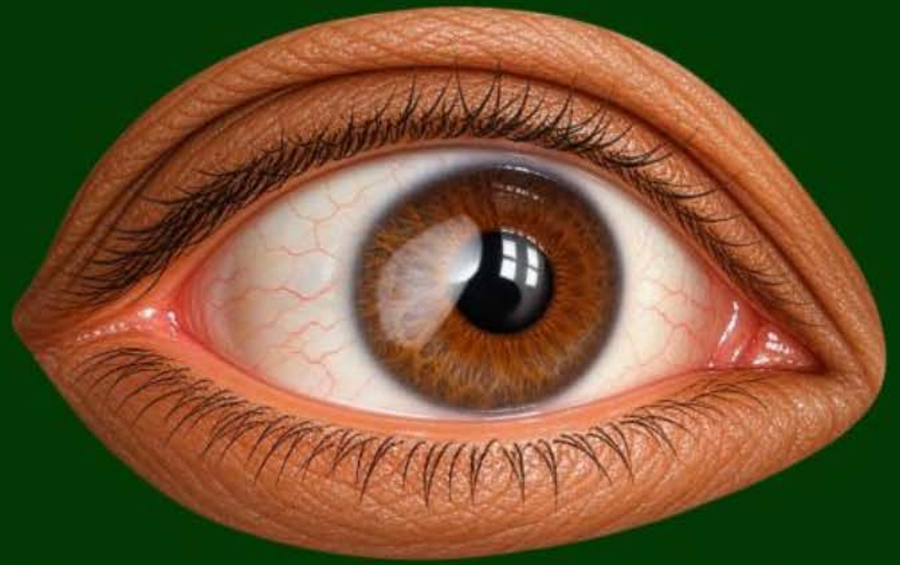
- It is a condition in which some people by birth cannot discriminate between certain colours such as red and green.
- This is due to a genetic defect. Mostly males suffer from this defect, whereas it rarely occurs in females.

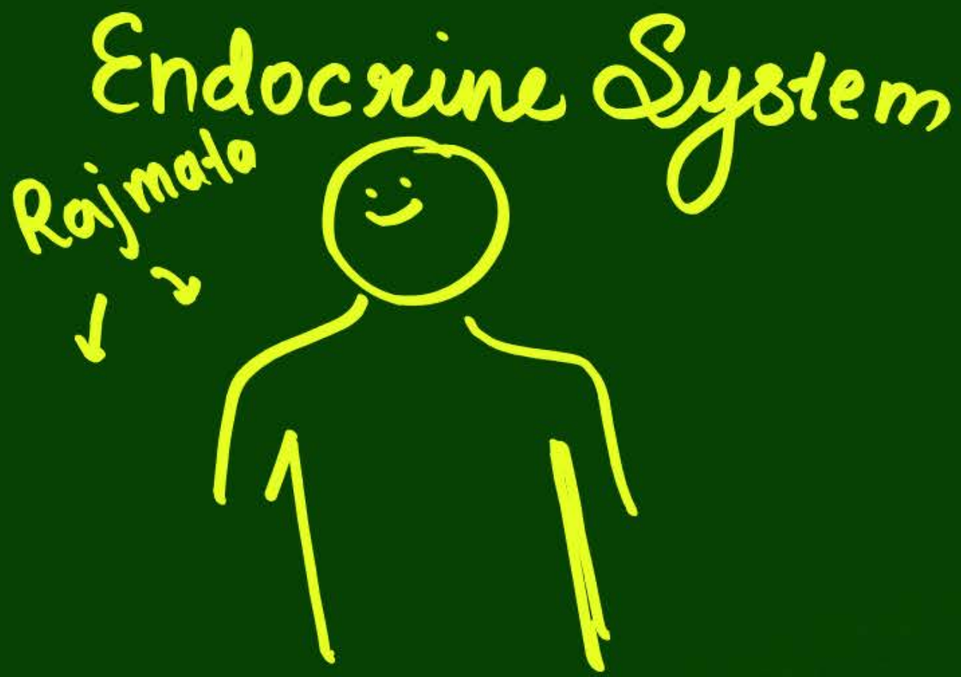
Sex-linked Recessive Disorder
(X)



CORNEAL OPACITIES

- The cornea of some patients gets scarred and turns opaque (white) and nonfunctional.
- Can cause anything from minor irritation to vision problems and even blindness.
- Treatment- the defective cornea can, be replaced by a healthy cornea from a donated eye.

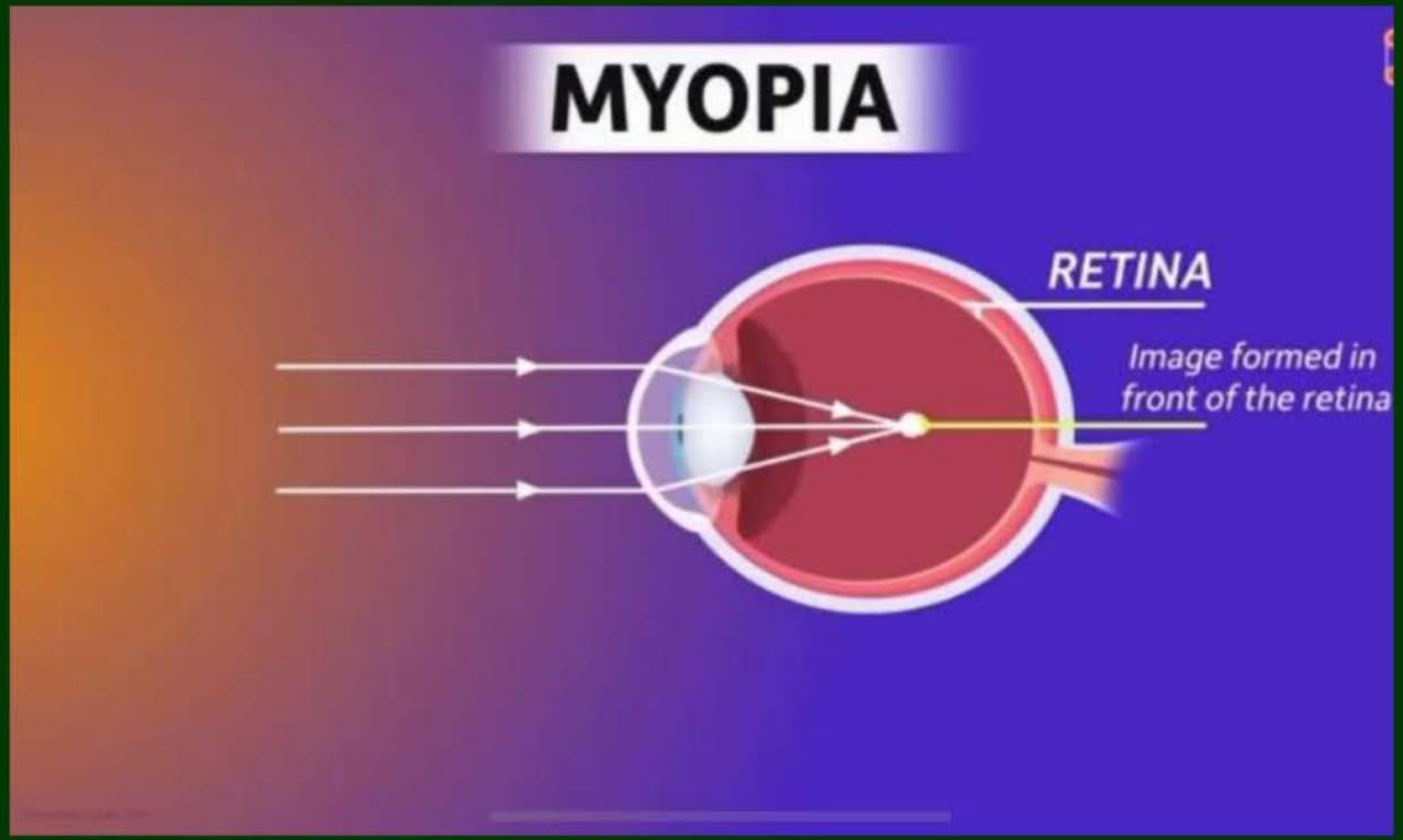


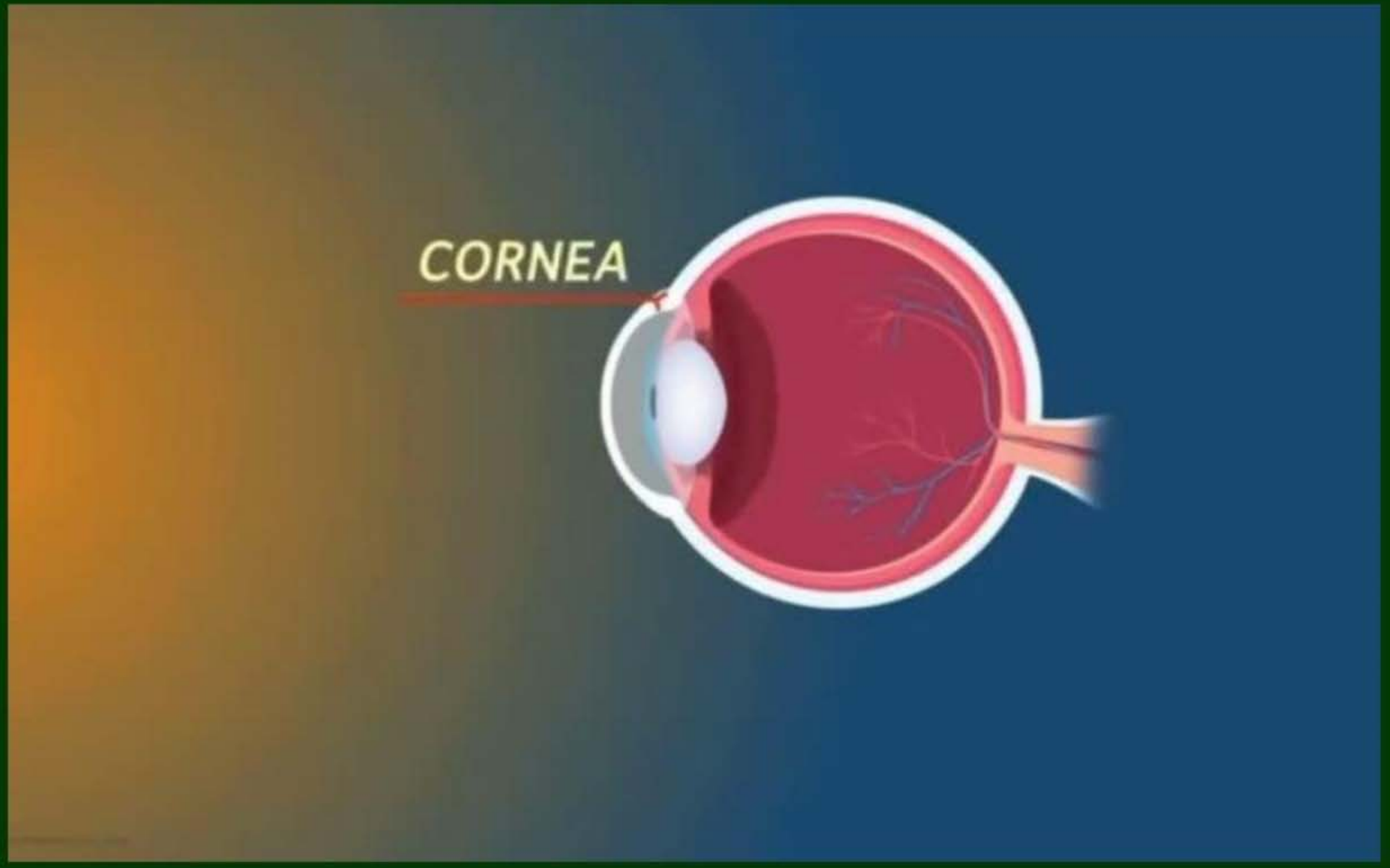

Endocrine System
Rajmata

SQUINT

- In this defect, the two eyes somewhat converge leading to what is called "cross eye".
- An opposite condition appears when they diverge and is called the "wide eye." Both conditions may cause double vision or diplopia.
- Treatment -Surgery and suitable exercise can correct these defects.




MYOPIA
RETINA
Image formed in
front of the retina


CORNEA

Thank You



